

Fermat's Little Theorem

First Version – If p is prime and a is an integer such that p does not divide a

$$a^{p-1} \equiv 1 \pmod{p}$$

Second Version

$$a^p \equiv a \pmod{p}$$

EXAMPLE

Find the result of $6^{10} \bmod 11$.

Solution

We have $6^{10} \bmod 11 = 1$. This is the first version of Fermat's little theorem where $p = 11$.

Find the result of $3^{12} \bmod 11$.

Solution

Here the exponent (12) and the modulus (11) are not the same. With substitution this can be solved using Fermat's little theorem.

$$3^{12} \bmod 11 = (3^{11} \times 3) \bmod 11 = (3^{11} \bmod 11) (3 \bmod 11) = (3 \times 3) \bmod 11 = 9$$

Multiplicative Inverses

$$a^{-1} \bmod p = a^{p-2} \bmod p$$

The answers to multiplicative inverses modulo a prime can be found without using the extended Euclidean algorithm:

- a. $8^{-1} \bmod 17 = 8^{17-2} \bmod 17 = 8^{15} \bmod 17 = 15 \bmod 17$
- b. $5^{-1} \bmod 23 = 5^{23-2} \bmod 23 = 5^{21} \bmod 23 = 14 \bmod 23$
- c. $60^{-1} \bmod 101 = 60^{101-2} \bmod 101 = 60^{99} \bmod 101 = 32 \bmod 101$
- d. $22^{-1} \bmod 211 = 22^{211-2} \bmod 211 = 22^{209} \bmod 211 = 48 \bmod 211$

Euler's Theorem

First Version-

if n and a are coprime positive integers

$$a^{\phi(n)} \equiv 1 \pmod{n}$$

Second Version

$$a^{k \times \phi(n) + 1} \equiv a \pmod{n}$$

The second version of Euler's theorem is used in the RSA cryptosystem

EXAMPLE

Find the result of $6^{24} \bmod 35$.

Solution

We have $6^{24} \bmod 35 = 6^{\phi(35)} \bmod 35 = 1$.

Find the result of $20^{62} \bmod 77$.

Solution

If we let $k = 1$ on the second version, we have

$$\begin{aligned} 20^{62} \bmod 77 &= (20 \bmod 77) (20^{\phi(77) + 1} \bmod 77) \bmod 77 \\ &= (20)(20) \bmod 77 = 15. \end{aligned}$$

Multiplicative Inverses

Multiplicative Inverses

Euler's theorem can be used to find multiplicative inverses modulo a composite.

$$a^{-1} \bmod n = a^{\phi(n)-1} \bmod n$$

EXAMPLE

The answers to multiplicative inverses modulo a composite can be found without using the extended Euclidean algorithm if we know the factorization of the composite:

- a. $8^{-1} \bmod 77 = 8^{\phi(77)-1} \bmod 77 = 8^{59} \bmod 77 = 29 \bmod 77$
- b. $7^{-1} \bmod 15 = 7^{\phi(15)-1} \bmod 15 = 7^7 \bmod 15 = 13 \bmod 15$
- c. $60^{-1} \bmod 187 = 60^{\phi(187)-1} \bmod 187 = 60^{159} \bmod 187 = 53 \bmod 187$
- d. $71^{-1} \bmod 100 = 71^{\phi(100)-1} \bmod 100 = 71^{39} \bmod 100 = 31 \bmod 100$

Mersenne Primes

$$\mathbf{M_p = 2^p - 1}$$

$$M_2 = 2^2 - 1 = 3$$

$$M_3 = 2^3 - 1 = 7$$

$$M_5 = 2^5 - 1 = 31$$

$$M_7 = 2^7 - 1 = 127$$

$$M_{11} = 2^{11} - 1 = 2047 \quad \text{Not a prime (2047 = 23 \times 89)}$$

$$M_{13} = 2^{13} - 1 = 8191$$

$$M_{17} = 2^{17} - 1 = 131071$$

A number in the form $M_p = 2^p - 1$ is called a Mersenne number and may or may not be a prime.

CHINESE REMAINDER THEOREM

The Chinese remainder theorem (CRT) is used to solve a set of congruent equations with one variable but different moduli, which are relatively prime, as shown below:

$$x \equiv a_1 \pmod{m_1}$$

$$x \equiv a_2 \pmod{m_2}$$

...

$$x \equiv a_k \pmod{m_k}$$

EXAMPLE

The following is an example of a set of equations with different moduli:

$$x \equiv 2 \pmod{3}$$

$$x \equiv 3 \pmod{5}$$

$$x \equiv 2 \pmod{7}$$

The solution to this set of equations is given in the next section; for the moment, note that the answer to this set of equations is $x = 23$. This value satisfies all equations: $23 \equiv 2 \pmod{3}$, $23 \equiv 3 \pmod{5}$, and $23 \equiv 2 \pmod{7}$.

STEPS

Solution To Chinese Remainder Theorem

1. Find $M = m_1 \times m_2 \times \dots \times m_k$. This is the common modulus.
2. Find $M_1 = M/m_1, M_2 = M/m_2, \dots, M_k = M/m_k$.
3. Find the multiplicative inverse of $M_1, M_2, \dots, M_k$ using the corresponding moduli $(m_1, m_2, \dots, m_k)$. Call the inverses $M_1^{-1}, M_2^{-1}, \dots, M_k^{-1}$.
4. The solution to the simultaneous equations is

$$x = (a_1 \times M_1 \times M_1^{-1} + a_2 \times M_2 \times M_2^{-1} + \dots + a_k \times M_k \times M_k^{-1}) \bmod M$$

EXAMPLE

Find the solution to the simultaneous equations:

Solution

We follow the four steps.

$$x \equiv 2 \pmod{3}$$

$$x \equiv 3 \pmod{5}$$

$$x \equiv 2 \pmod{7}$$

1. $M = 3 \times 5 \times 7 = 105$

2. $M_1 = 105 / 3 = 35$, $M_2 = 105 / 5 = 21$, $M_3 = 105 / 7 = 15$

3. The inverses are $M_1^{-1} = 2$, $M_2^{-1} = 1$, $M_3^{-1} = 1$

4. $x = (2 \times 35 \times 2 + 3 \times 21 \times 1 + 2 \times 15 \times 1) \bmod 105 = 23$
 $\bmod 105$

EXAMPLE: CRT

1. A group of people are arranging themselves for a parade. If they line up three to a row, one person is left over. If they line up four to a row, two people are left over, and if they line up five to a row, three people are left over. What is the smallest possible number of people?